

Noise cannot help quantum-enhanced MCMC: exactness is free, advantage is not

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Quantum-walk Metropolis proposals give an exactly correct Markov-chain sampler whenever the proposal is symmetric, without evaluating proposal probabilities. We classify the noise channels that preserve that free lunch: every transpose-closed Kraus set — dephasing in any real basis, depolarizing noise, and equal-rate raising and lowering — yields a symmetric proposal at any strength. The condition extends to Lindblad semigroups with a real-symmetric walk Hamiltonian by a time-reversed Dyson unravelling. Amplitude damping (T_1) is the unique realistic violator; a device that keeps the evaluation-free Metropolis ratio on a damped proposal runs naive Metropolis rather than Hastings, with a proved per-step total-variation bound of order the symmetry defect and a (typically loose) stationary envelope. Exactness is therefore free for the noise that dominates today’s devices. Advantage is not. Intermediate dephasing can accelerate the walk’s own mixing — the interior optimum is a known property of decohered walks, and we give an exact window-averaged telegraph solution that isolates the even-moment coherences time randomization cannot kill. Those Goldilocks peaks never lift the sampler above the envelope of the coherent quench and trivial classical kernels on Gibbs targets. A pre-registered test on fifty random Ising, Sherrington–Kirkpatrick, and random-field instances finds the envelope law in forty-nine cells, with one disclosed +1.17% instance-specific exception; in every cell where the coherent quench already wins, dephasing erodes the win. The Metropolis filter is why environment-assisted transport does not transfer to sampling: delocalization broadens proposal energy, and the filter rejects the spread. Finally we test what the coherent quench’s value actually is, against a fourteen-kernel exact-tier zoo: energy concentration alone does not predict advantage (informed local kernels concentrate best and mix worst), but concentration *at Hamming distance* does, and an oracle that merely proposes uniformly from the target’s energy shell matches or beats the quench everywhere — a mechanism no cheap classical kernel we construct reproduces, and whose exact classical emulation pays a measured certification cost of at least one order of magnitude every four spins, because an exact shell proposal must certify a near-empty energy window complete — a proof a physical sampler never performs. That asymmetry generalizes: a conserved diagonal charge confines every measured outcome to its symmetry sector while preserving evaluation-free exactness within it, a feasibility certificate priced at zero; but a pre-registered race of the amortized architecture this suggests — device-mined proposal libraries consumed by exact mode-jumping — is killed by its own criteria, classically. Noise costs nothing and buys nothing; the quench’s edge is global isoenergetic proposals.

I. INTRODUCTION

Markov-chain Monte Carlo (MCMC) draws samples from a target π on a set too large to enumerate by proposing a move and accepting or rejecting it. The quality of the sampler is the quality of the proposals. Layden *et al.* [1] showed that a short coherent quantum quench, used only as a proposal inside an otherwise classical Metropolis–Hastings (MH) kernel, can suggest long-range, energy-preserving jumps that a local classical kernel misses, and that the construction remains *exactly* correct whenever the proposal is symmetric — no evaluation of the proposal probabilities is required. On hardware they observed that the sampler stayed accurate under realistic noise. That robustness was treated as a nuisance property, stated in their Supplemental Material as a symmetry-on-average condition, and left unclassified.

A different literature makes noise look like a resource. In environment-assisted quantum transport (ENAQT), a

moderate dephasing rate delocalizes an excitation that pure coherent dynamics would trap, and a large rate freezes it [2, 3]. The same “Goldilocks” interior optimum appears in the mixing time of decohered quantum walks on the line, the cycle, and the hypercube [4–7] — a phenomenon we cite as known, and return to in Secs. IV and VIII. The tempting conjecture for today’s noisy devices is that a tuned dephasing rate should likewise help a quantum-walk *Metropolis proposal*: noise would cost nothing (exactness already survived it in Layden’s experiment) and might buy mixing (ENAQT and the walk literature say it can). We call that conjecture noise-as-resource, and we tested it as three hypotheses: H1, a Goldilocks peak in sampling efficiency; H2, a stable matching γ^*/Δ across targets; H3, a hardware walk that beats its own zero-idle configuration because native noise already sits near γ^* .

The answer is two-sided, and we do not flatten it. Realistic decoherence splits cleanly. Every transpose-closed noise channel — dephasing in any real basis, depolarizing noise, infinite-temperature relaxation — preserves the evaluation-free exactness of the sampler at *any* strength. T_1 amplitude damping is the unique realistic channel that does not, and the bias it induces when a device keeps the

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free-lunch Metropolis ratio is bounded. That is the positive half: exactness is free. The negative half is not “noise never helps mixing.” Noise *does* create Goldilocks peaks. In a solvable corner the window-averaged gap rises from 0.5 to 0.97; in the field, twenty of fifty random cells show an interior peak. Those peaks never lift the sampler above the envelope of the coherent quench and the trivial classical kernels. Wherever the quantum proposal already has an advantage, dephasing erodes it; wherever dephasing helps the walk, a classical kernel was already better. Advantage is not free, and H1–H3 die in that form. Noise costs nothing and buys nothing.

We make six contributions.

1. *A characterization of exactness-preserving noise* (Theorems 1 and 2). Transpose-closed Kraus sets give symmetric proposals; the condition extends to Lindblad semigroups with real-symmetric H and transpose-closed jumps via a time-reversed Dyson unravelling. Layden’s SM stated the symmetry-on-average condition; we classify the channels that satisfy it. Certification places every transpose-closed class at defect $\leq 1.2 \times 10^{-14}$ and the violators at $\geq 3.5 \times 10^{-2}$.
2. *A quantitative T_1 budget* (Theorems 3 and 4). A device that ignores the q -ratio on a damped proposal runs naive Metropolis, not Hastings. The per-step total-variation distance is at most $\varepsilon(1/\pi_{\min} - 1)$; the stationary distance is at most that times $\pi_{\min}^{-1/2}/\delta$. Measured TV is $O(\varepsilon)$; the gap envelope is guaranteed and typically loose by 10^6 – 10^{21} . The working device requirement is $\varepsilon \lesssim \eta$, or else estimate the q -ratio.
3. *A refutation of gap-monotonicity, a solvable corner, and the envelope law* (Sec. IV). Gap-monotonicity in γ is false. The window-averaged telegraph solution plus the odd/even-moment mechanism explain *why* noise can help mixing, and embed that known phenomenon in an exact MH sampler. The envelope law — noise never beats max(coherent, classical) — is pre-registered and holds in 49/50 random cells, with one disclosed +1.17% instance-specific exception. In advantage cells, dephasing erodes beyond $\gamma \approx 0.03\Delta$. Orfi and Sels already rule out a worst-case unital speedup [8]; our law is the per-instance, family-wide complement.
4. *The quench’s mechanism, measured against its classical competition* (Secs. VII and VIIA). The scalar energy-concentration conjecture is refuted by a fourteen-kernel exact-tier zoo; concentration *at Hamming distance* survives as the predictor; an oracle that merely proposes uniformly from the energy shell matches or beats the quench everywhere it is connected; and three classical routes to that shell are measured and fall short — informed local

kernels (monotonically in their informedness), two-replica isoenergetic cluster moves (never above the mixture-dilution floor), and de novo shell sampling by SAT and pseudo-Boolean solvers (a wall-clock gap to the quench that grows from about three orders at $n = 16$ – 20 to at least four and a half by $n = 24$, because an exact proposal must certify a near-empty energy window complete).

5. *A reusable benchmarking protocol* (Sec. VI). Kernels never see target log-probabilities; ESS per step and per second are both reported; split- $\hat{R} \leq 1.05$ gates frozen chains; kill criteria are written before data; negative results are reported at full strength. The protocol killed H1 on the G01 pilot and then caught the overreach of conjecture v1, forcing the G05 pre-registration.
6. *Extraction pricing as a design rule, with a free-certificate theorem and a registered negative* (Secs. IIIE and VIA). Devices charge per measured shot (10^2 – 10^4 s $^{-1}$) while classical samplers execute 10^6 – 10^8 proposals per second, so any hybrid that consumes device output per MCMC step starts 10^4 – $10^6\times$ behind on targets whose π -evaluation is cheap relative to shot latency. What noisy hardware does give away is invariants: a conserved diagonal charge confines every outcome to its symmetry sector (a feasibility certificate at zero cost, Theorem 5) while preserving the evaluation-free symmetry of Sec. III within each sector. We tested the one architecture that takes the device off the per-step path — a fixed atom library mined offline, consumed by exact mode-jumping proposals — under pre-registered kill criteria (run P01), and both kills fired: classical miners matched the simulated quench’s mode coverage in every cell at matched budget, and the darting gain never reached $2\times$ over parallel tempering on metastability-screened targets at $n = 50$ – 200 . The negative is reported at full strength.

Section II fixes notation. Section III is exactness. Section IV is advantage. Section V is the mechanism that stops ENAQT from transferring to sampling — the Metropolis filter, not a failure of delocalization. Section VII states the forward-looking predictor, energy concentration, as a conjecture — and then tests it on a fourteen-kernel exact-tier zoo (Sec. VIIA): the scalar fails, concentration *at distance* holds, and a shell oracle shows the mechanism is sufficient. Related work is collected in Sec. VIII rather than here; the short version is that we are the first channel-level classification of exactness for noisy walk proposals, and the first embedding of decohered-walk mixing results in an exact MH sampler with Gibbs targets.

For quantum-walk Metropolis proposals we (i) characterize exactly which noise channels preserve evaluation-free exactness (transpose-closed Kraus sets — dephasing in any real basis, depolarizing, equal-rate $\sigma^\pm$ — at

any strength, with T_1 the unique realistic bias channel), and (ii) establish the envelope law: tuned dephasing, although it genuinely accelerates the walk's own mixing as known since Kendon and Tregenna [4], never lifts the sampler's efficiency above the envelope of the coherent quench and trivial classical kernels on Gibbs targets. Exactness is free, advantage is not.

II. SETUP

We sample a target π on the hypercube $\{0,1\}^n$. The Metropolis–Hastings (MH) kernel built from a proposal $q(\cdot|x)$ is

$$P(y|x) = q(y|x) a(y, x) \quad (y \neq x), \quad (1)$$

with $P(x|x) = 1 - \sum_{y \neq x} P(y|x)$ and acceptance

$$a(y, x) = \min\left(1, \frac{\pi(y) q(x|y)}{\pi(x) q(y|x)}\right). \quad (2)$$

If q is symmetric, $q(y|x) = q(x|y)$, the proposal ratio cancels and the acceptance collapses to the evaluation-free Metropolis ratio $\min(1, \pi(y)/\pi(x))$. That is the free lunch of Layden *et al.* [1]: a quantum proposal that never has to be evaluated still yields an exactly correct sampler, provided symmetry holds.

The family we study is a *dephased quantum walk* used only as a proposal. The walk Hamiltonian is real symmetric,

$$H = (1 - \alpha) \text{diag}(E) + \alpha \sum_{i=1}^n X_i, \quad (3)$$

with mixer weight $\alpha \in (0, 1]$ and E the classical energy whose Gibbs distribution is π . The generator is the Lindbladian

$$\begin{aligned} \mathcal{L}(\rho) &= -i[H, \rho] + \gamma \sum_{i=1}^n \mathcal{D}[Z_i](\rho), \\ \mathcal{D}[A](\rho) &= A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}. \end{aligned} \quad (4)$$

A single proposal from x is a computational-basis measurement of $\mathcal{E}_t(|x\rangle\langle x|)$,

$$q_t(y|x) = \langle y | \mathcal{E}_t(|x\rangle\langle x|) | y \rangle, \quad \mathcal{E}_t = e^{t\mathcal{L}}. \quad (5)$$

Following Layden, we average t uniformly over a fixed window (in units of the spectral width Δ of H ; the G01/G05 window is $t\Delta \in [2, 12]$). The window-averaged proposal is written q_γ . Two limits are named: $\gamma = 0$ is the coherent quench of Layden *et al.*; $\gamma \rightarrow \infty$ is a Zeno freeze in the computational basis, recovered in practice as a lazy classical kernel.

Two equivalent realizations of $\mathcal{E}_t$ are used throughout. At $n \leq 6$ we exponentiate the dense superoperator (or apply it sparsely on a time grid) and read the

proposal matrix exactly. At larger n the same channel is sampled by a quantum-trajectory unravelling: one shot is one Poisson record of phase kicks, one trajectory, and one proposal. That unravelling is also the circuit recipe (stochastic Hamiltonian phase kicks, or a collision scheme with ancilla resets). Because the jump statistics of computational-basis dephasing are state-independent, the trajectory sampler never consults π .

Classical baselines, used only as competitors and never as ingredients of the walk, are Metropolized single-flip (Hamming-1) and uniform-flip (the complete graph). Efficiency is reported as the MH spectral gap $\delta(P) = 1 - |\lambda_2(P)|$ wherever the $2^n \times 2^n$ kernel can be built, and as effective sample size per step and per second on gated MCMC runs (Sec. VI). Nothing in this section is a claim about noise; the claims begin when $\gamma > 0$ or when the channel is replaced by a different physical noise model.

III. EXACTNESS IS FREE

The free lunch requires $q(y|x) = \langle y | \mathcal{E}(|x\rangle\langle x|) | y \rangle$ to be symmetric in (x, y) . Layden *et al.* already state the operational condition in their Supplemental Material: errors are harmless “provided such errors do not break the $Q(s'|s) = Q(s|s')$ symmetry on average.” Our contribution is the channel-level *classification* of which physical noise satisfies that condition, at any strength, together with a quantitative bound for the one realistic violator.

The proposal basis $\{|x\rangle\}$ is the computational basis throughout; transposes are taken in this basis.

A. Transpose-closed channels

Theorem 1 (Transpose-closed Kraus sets) *Let $\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$ admit a Kraus representation $\{K_a\}_{a=1}^m$ that is closed under transposition up to unitary remixing,*

$$K_a^\top = \sum_b u_{ab} K_b \quad \text{for some unitary } u \in \text{U}(m). \quad (\text{TC})$$

Then $q(y|x) = q(x|y)$ for all x, y .

Proof. For real basis vectors, $\langle x | K_a | y \rangle = \langle y | K_a^\top | x \rangle$. Hence

$$\begin{aligned} q(x|y) &= \sum_a |\langle x | K_a | y \rangle|^2 = \sum_a |\langle y | K_a^\top | x \rangle|^2 \\ &= \sum_a \left| \sum_b u_{ab} \langle y | K_b | x \rangle \right|^2 = \sum_b |\langle y | K_b | x \rangle|^2 = q(y|x), \end{aligned} \quad (6)$$

the last-but-one step because u preserves the ℓ^2 -norm of the amplitude vector $v_b = \langle y | K_b | x \rangle$. $\square$

Condition (TC) says that $\{K_a^\top\}$ is *also* a Kraus representation of $\mathcal{E}$. The set $\{K_a^\top\}$ always represents the map

$\Theta \circ \mathcal{E}^* \circ \Theta$, where Θ is transposition and $\mathcal{E}^*$ is the Heisenberg adjoint (Kraus $\{K_a^\dagger\}$). Two Kraus sets represent the same channel if and only if they are unitarily related, so (TC) is exactly the representation-independent identity

$$\mathcal{E} = \Theta \circ \mathcal{E}^* \circ \Theta. \quad (\text{TC}')$$

A channel either has a transpose-closed Kraus set or it does not.

Unitary remixing preserves $\sum_a A_a^\dagger A_a$, so (TC) implies $\sum_a (K_a^\dagger)^\dagger (K_a^\dagger) = \sum_a K_a^\dagger K_a = I$, hence $\mathcal{E}(I) = I$: every transpose-closed channel is unital. A symmetric proposal is column-stochastic and symmetric, hence doubly stochastic, and $\sum_x q(y|x) = \langle y | \mathcal{E}(I) | y \rangle$, so

$$\text{diag}(\mathcal{E}(I)) = 1 \quad \text{is necessary for symmetry.} \quad (\text{N})$$

Non-unital noise with a non-flat diagonal of $\mathcal{E}(I)$ — any channel that relaxes toward a preferred state, including finite-temperature amplitude damping — breaks symmetry before any interference argument is needed.

Condition (TC) is sufficient, not necessary. Symmetry constrains only the diagonal-to-diagonal block of $\mathcal{E}$ (the transition matrix $P = P^\top$), while (TC') constrains the whole superoperator. Contrived channels can be symmetric without (TC). The honest statement, and the one we make, is structural: every physically arising symmetric case below satisfies (TC), and the physically arising violator (finite- T damping) already fails the weaker necessary condition (N).

B. Lindblad semigroups

Theorem 2 (Lindblad semigroups) *Let $\mathcal{L}(\rho) = -i[H, \rho] + \sum_j \gamma_j \mathcal{D}[L_j](\rho)$. Suppose (i) $H = H^\top$ is real symmetric; (ii) the weighted jump set is transpose-closed: there is a permutation $j \mapsto j'$ with $L_j^\top = e^{i\phi_j} L_{j'}$ and $\gamma_j = \gamma_{j'}$; (iii) $\sum_j \gamma_j L_j^\dagger L_j$ is real symmetric, or equivalently the effective non-Hermitian Hamiltonian $H_{\text{eff}} = H - (i/2) \sum_j \gamma_j L_j^\dagger L_j$ satisfies $H_{\text{eff}}^\top = H_{\text{eff}}$. Then $\mathcal{E}_t = e^{t\mathcal{L}}$ satisfies (TC) for every $t \geq 0$, hence $q_t(y|x) = q_t(x|y)$ for all t .*

Proof (time-reversed Dyson unravelling). The time-ordered decomposition of $e^{t\mathcal{L}}$ is the continuous Kraus family indexed by jump records $\omega = (m; t_1 < \dots < t_m; j_1, \dots, j_m)$,

$$K_\omega = G(t - t_m) \sqrt{\gamma_{j_m}} L_{j_m} \cdots \sqrt{\gamma_{j_1}} L_{j_1} G(t_1), \quad G(s) = e^{-iH_{\text{eff}} s}, \quad (7)$$

with $e^{t\mathcal{L}}(\rho) = \sum_m \int dt_1 \cdots dt_m \sum_{j_1 \dots j_m} K_\omega \rho K_\omega^\dagger$. Transposition reverses operator order. By (iii), $G(s)^\top = G(s)$; by (ii), each $L_j^\top$ is a phase times $L_{j'}$ at the same rate. Hence

$$K_\omega^\top = e^{i\Phi(\omega)} K_{\tilde{\omega}}, \quad \tilde{\omega} = (m; t - t_m < \dots < t - t_1; j'_m, \dots, j'_1), \quad (8)$$

the time-reversed, transpose-relabelled record. The map $\omega \mapsto \tilde{\omega}$ is a measure-preserving involution of the index set (Lebesgue measure on the time-simplex is invariant under $s \mapsto t - s$; the rate weights match by (ii)), and phases are killed by $|\cdot|^2$. The Kraus family is therefore transpose-closed and Theorem 1 applies. $\square$

Condition (iii) is implied by (i)+(ii) when the jumps are real (up to phase) and normal ($L^\top L = LL^\top$); every row of Table I has normal jumps. Time-reversal is load-bearing: a product of symmetric operators is not symmetric, $(AB)^\top = BA$, so “each instant is symmetric” does not give “the semigroup is symmetric.” Naive composition fails; the Dyson involution is the mechanism. Symmetry survives convex mixtures — including Layden’s random t -window and a random γ — and time-reversal-symmetric compositions, but not arbitrary composition of two different symmetric channels.

C. Which physical noise is free

Table I and Fig. 1 collect the classification. Computational-basis dephasing (Z_i jumps, any rate), dephasing in *any* real basis (jumps $A = VDV^\top$ with V real orthogonal), depolarizing and arbitrary Pauli-jump noise, and infinite-temperature relaxation (σ^- and σ^+ at equal rates) all satisfy (TC). Amplitude damping (σ^- only) and finite- T relaxation at unequal rates fail (N) and break symmetry.

The last two rows are the sharp pair: the same jump operators, and the matching condition $\gamma_+ = \gamma_-$ is exactly what flips symmetry on. Noise that drives toward the maximally mixed state preserves the free lunch; noise that knows a preferred state (T_1 relaxation toward $|0\rangle$) breaks it.

Numerical certification pins every row. The worst defect over all transpose-closed channels is 1.2×10^{-14} across $n \in \{4, 5\}$ ($n = 6$ for computational dephasing), $\gamma/\Delta \in \{0.1, 1, 10\}$, and $t\Delta \in \{0.5, 2, 8\}$. Both negative controls — amplitude damping, and unequal-rate $\sigma^\pm$ with $\gamma_+ = \gamma_-/4$ — break at defect $\geq 3.5 \times 10^{-2}$. The certified-file maximum for amplitude damping is 0.352 (at $\kappa = 0.1\Delta$, $t\Delta \leq 8$); an older grid’s 0.55 is not in the current file and is not used here.

Device noise is, to a first accounting, dephasing (T_φ) plus amplitude damping (T_1) plus gate depolarizing plus coherent overrotation (real H errors). By Table I, T_φ at any rate, depolarizing gate noise, and real-Hamiltonian miscalibration are all exactness-free. T_1 relaxation is the *unique realistic bias channel*. A hardware claim has to bound or estimate only the damping component; everything else rides free. That is the formal content of the robustness Layden *et al.* observed empirically.

TABLE I. Channel classes, transpose closure, and certified symmetry defect $\max_{x,y} |q(y|x) - q(x|y)|$.

Noise channel	Transpose closure	Symmetric?	Certified defect
Computational-basis dephasing, jumps Z_i , any rate	$Z_i^\top = Z_i$	yes	$\leq 3.3 \times 10^{-16}$
Dephasing in any real basis, $A = VDV^\top$	$A^\top = A$	yes	$\leq 1.3 \times 10^{-14}$
Depolarizing / Pauli-jump noise (rates arbitrary)	$X^\top = X, Z^\top = Z, Y^\top = -Y$	yes	$\leq 10^{-15}$
Infinite- T relaxation (equal-rate $\sigma^\pm$)	$(\sigma^-)^\top = \sigma^+$, rates matched	yes	$\leq 10^{-15}$
Amplitude damping / unequal-rate $\sigma^\pm$	$(\sigma^-)^\top = \sigma^+ \notin \text{set}$; fails (N)	no	$\geq 3.5 \times 10^{-2}$ (file max 0.352)

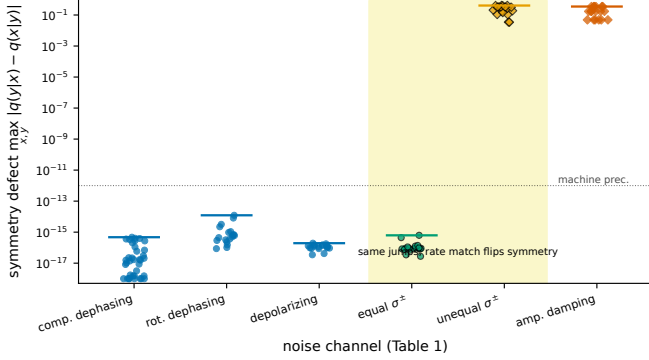


FIG. 1. Symmetry defect versus channel class on a log scale. Transpose-closed classes sit at machine precision; amplitude damping and unequal-rate $\sigma^\pm$ sit between 3.5×10^{-2} and 0.352. The equal/unequal $\sigma^\pm$ pair is highlighted: the same jumps, and only the rate-matching condition, decide exactness.

D. T_1 bias: naive Metropolis versus Hastings

A device that keeps the free-lunch Metropolis ratio on a κ -damped proposal $\tilde{q}$ is *not* running the exact Hastings kernel. Write P_H for Hastings on $\tilde{q}$ (acceptance includes the q -ratio; stationary distribution exactly π) and P_M for naive Metropolis (acceptance $\min(1, \pi(y)/\pi(x))$; stationary $\tilde{\pi} \neq \pi$ in general). The theorem is not $\|\tilde{q}_{\kappa=0} - \tilde{q}_\kappa\|$; that is a proposal-TV statement and is not controlled by the symmetry defect. Write $\varepsilon = \max_{x,y} |\tilde{q}(y|x) - \tilde{q}(x|y)|$ for the symmetry defect.

Theorem 3 (Per-step TV) *For every state x ,*

$$\begin{aligned} \|P_M(\cdot|x) - P_H(\cdot|x)\|_{\text{TV}} &\leq \sum_{y \neq x} \frac{\pi(y)}{\pi(x)} |\tilde{q}(y|x) - \tilde{q}(x|y)| \\ &\leq \varepsilon(1/\pi(x) - 1). \end{aligned} \quad (9)$$

Uniformly over x ,

$$\|P_M - P_H\|_{\text{TV},\infty} \leq \varepsilon(1/\pi_{\min} - 1), \quad (10)$$

where $\pi_{\min} = \min_x \pi(x)$ and $\|\cdot\|_{\text{TV},\infty} = \max_x \|\cdot|x\|_{\text{TV}}$.

Proof. For $y \neq x$, $|P_M(y|x) - P_H(y|x)| = \tilde{q}(y|x) |\min(1, r) - \min(1, r\rho)|$ with $r = \pi(y)/\pi(x)$ and $\rho = \tilde{q}(x|y)/\tilde{q}(y|x)$ ($\rho := 1$ if $\tilde{q}(y|x) = 0$). The

elementary inequality $|\min(1, a) - \min(1, b)| \leq |a - b|$ gives $|\min(1, r) - \min(1, r\rho)| \leq r|1 - \rho|$, hence

$$|\Delta P(y|x)| \leq \tilde{q}(y|x) r |1 - \rho| = \frac{\pi(y)}{\pi(x)} |\tilde{q}(y|x) - \tilde{q}(x|y)|. \quad (11)$$

The diagonal difference equals $-\sum_{y \neq x} \Delta P(y|x)$, so $|\Delta P(x|x)| \leq \sum_{y \neq x} |\Delta P(y|x)|$. Therefore

$$\begin{aligned} \|\Delta P(\cdot|x)\|_{\text{TV}} &= \frac{1}{2} \sum_y |\Delta P(y|x)| \\ &\leq \sum_{y \neq x} |\Delta P(y|x)| \leq \varepsilon(1/\pi(x) - 1). \end{aligned} \quad (12)$$

□

If $\varepsilon = 0$ — every transpose-closed channel in Table I — then $P_M = P_H$ and there is no free-lunch bias. T_1 remains the unique realistic channel that can make $\varepsilon > 0$. Hastings on a T_1 -damped proposal is still exact; the cost is evaluating the q -ratio.

Theorem 4 (Stationary TV) *Let $\delta = \delta(P_H) = 1 - |\lambda_2(P_H)|$ be the spectral gap of the exact Hastings kernel. Then*

$$\begin{aligned} \|\tilde{\pi} - \pi\|_{\text{TV}} &\leq \frac{1}{\delta} \pi_{\min}^{-1/2} \|P_M - P_H\|_{\text{TV},\infty} \\ &\leq \varepsilon(1/\pi_{\min} - 1) \pi_{\min}^{-1/2} / \delta. \end{aligned} \quad (13)$$

Proof. Stationarity gives $\tilde{\pi} P_M = \tilde{\pi}$ and $\pi P_H = \pi$, so $(\tilde{\pi} - \pi)(I - P_H) = \tilde{\pi}(P_M - P_H) =: \nu$. Work in $L^2(\pi)$ via densities $f = d\nu/d\pi$. P_H is reversible, hence self-adjoint on densities in $L^2(\pi)$; on the mean-zero subspace, $\|(I - P_H)^{-1}\|_{L^2(\pi)} \leq 1/\delta$. Both $\tilde{\pi} - \pi$ and ν are mean-zero. With $f = d(\tilde{\pi} - \pi)/d\pi$ and $g = d\nu/d\pi$,

$$\|\tilde{\pi} - \pi\|_{\text{TV}} = \frac{1}{2} \|f\|_{L^1(\pi)} \leq \frac{1}{2} \|f\|_{L^2(\pi)} \leq \frac{1}{2\delta} \|g\|_{L^2(\pi)}. \quad (14)$$

Then $\|g\|_{L^2(\pi)}^2 = \sum_y \nu(y)^2 / \pi(y) \leq \pi_{\min}^{-1} (\sum_y |\nu(y)|)^2$, so $\|g\|_{L^2(\pi)} \leq \pi_{\min}^{-1/2} \cdot 2\|\nu\|_{\text{TV}}$. Finally $\|\nu\|_{\text{TV}} = \|\tilde{\pi}(P_M - P_H)\|_{\text{TV}} \leq \|P_M - P_H\|_{\text{TV},\infty}$. Combining and applying Theorem 3 gives the claim. □

The simpler expression $B_{\text{naive}} = \varepsilon(1/\pi_{\min} - 1)/\delta$ — Theorem 4 without the $\pi_{\min}^{-1/2}$ factor — is *not* proved. The $1/\delta$ inverse lives in $L^2(\pi)$, and the passage to TV

costs $\pi_{\min}^{-1/2} \cdot B_{\text{naive}}$ held in every certified cell (with 10^6 – 10^{21} slack); we treat it as empirical. The theorem is the displayed line.

The factor $\pi_{\min}^{-3/2}/\delta$ is exponentially pessimistic at low T : $\pi_{\min}$ is Gibbs-small and δ is the worst-case gap. This is a guaranteed envelope, not a prediction. Certification ($n \leq 5$, three e01-chain cells, five κ/Δ values) finds measured stationary TV of order ε , typically 0.7 – $3 \times$ the per-step TV, while the envelope sits 10^6 – 10^{21} above it. At $T = 0.3$ the uniform bound is vacuous ($\pi_{\min} \sim 10^{-18}$) while the measured bias at device-relevant $\kappa/\Delta = 0.01$ is 2.5%. No claim is made at hardware n ; no claim is made that T_1 is negligible on devices. The claim is: T_1 is the only realistic term to budget, the per-step TV error is $O(\varepsilon)$, and ε is a few percent at $\kappa \sim 0.01\Delta$ on the G01 window (tens of percent at $\kappa \sim 0.1\Delta$).

A device that needs $\|\tilde{\pi} - \pi\|_{\text{TV}} < \eta$ should treat $\varepsilon \lesssim \eta$ as the working requirement — what the numbers do — not $\varepsilon \lesssim \eta \delta \pi_{\min}$, what the envelope says. If that is too tight for the device's T_1 , estimate the q -ratio and run Hastings.

E. Free certificates from conserved charges

The exactness results above have a feasibility-side companion. Let $q : \{0,1\}^n \rightarrow \mathbb{R}$ be a classical constraint function and $Q = \sum_x q(x)|x\rangle\langle x|$ the corresponding diagonal charge, so that each Q -eigenspace is spanned by computational-basis states.

Theorem 5 (Sector support) *Prepare $|x\rangle$, apply any finite composition of unitaries U with $[U, Q] = 0$ and CPTP maps whose every Kraus operator commutes with Q , and measure in the computational basis. Then every outcome y satisfies $q(y) = q(x)$ with probability 1, at any depth, evolution time, and system size. If additionally the hypotheses of Theorem 2 hold (real symmetric H , transpose-closed jumps), the proposal is symmetric within every sector simultaneously, hence a valid $\log q \equiv 0$ MH proposal for any target restricted to the start sector.*

The proof is commutant algebra plus the time-reversed unravelling of Theorem 2, sector-restricted; commuting operators are block-diagonal across Q -eigenspaces, so sector membership is preserved by induction over the composition, and the symmetry argument is unchanged (proof and a machine certification — $n = 6$ number-conserving XY dynamics with diagonal disorder under full-Lindblad dephasing; cross-sector leakage exactly zero, symmetry defect 1.9×10^{-16} — are distributed with the manuscript source). Two readings matter here. First, every diagonal jump operator commutes with every diagonal Q , so computational-basis dephasing — the dominant channel, and the free one by Theorem 2 — is automatically sector-preserving: for diagonal charges, dephasing costs neither exactness nor feasibility. Second, the theorem prices at zero exactly what

Sec. VII A measures as the scaling cost of every classical route to the energy shell: a classical exact proposal must *certify* its constraint set (the completeness proofs censored at $n \geq 24$), while conserving dynamics cannot leave it. The ingredients are individually standard — decoherence-free subspaces [9], symmetry-verification error mitigation [10], conserving drivers for constrained annealing and QAOA [11, 12] — but they have been used to protect or mitigate an output-priced computation; here the sector *is* the encoding, for sampling, inside an exact MH boundary. Hamming-weight sectors are the natural hardware instantiation (XY/iSWAP gate sets conserve particle number exactly). What the theorem does not give: any statement under amplitude damping. Discarding sector-violating shots (a classical $O(n^2)$ check per shot) restores the support guarantee by fiat, and we conjecture it also preserves proposal symmetry — but the discard probability is state-dependent for exactly the reason T_1 breaks Theorem 1, so this remains a conjecture, not a corollary, and we do not use it. Fair sampling *within* a sector is likewise not implied (annealers are measurably unfair samplers of degenerate sets [13]); it is an empirical question, and Sec. VI A is what measuring it honestly looks like.

IV. ADVANTAGE IS NOT

Exactness is free for every transpose-closed channel. Speed is not. The natural first conjecture after the G01 pilot — that the MH gap $\delta(\gamma)$ is monotone non-increasing in the dephasing rate for every target — is false. Noise really does create Goldilocks peaks in the walk's own mixing. The peaks never lift the sampler above the envelope of the coherent quench and the trivial classical kernels. That is the envelope law: proved nowhere, pre-registered and tested on fifty fresh cells, and complementary to the worst-case unital bound of Orfi and Sels [8].

A. A refuted monotonicity

The G01 pilot ($n = 6$, $\alpha \leq 0.5$, $T \leq 1$; Sec. IV D) found $\delta(\gamma)$ monotone decreasing in all twelve cells. The sketch offered for that pattern treated the dephased kernel as a mixture and invoked a data-processing intuition: mixing kernels cannot beat their best member. Both halves of the sketch fail.

Lemma 1 (Mixture structure) *The window-averaged dephased proposal q_γ produces an MH transition matrix that is an exact convex mixture $T_\gamma = \mathbb{E}_{t,\omega}[T_{t,\omega}]$, where ω is the Poisson($n\gamma$) phase-kick record, $T_{t,\omega}$ is the MH chain built from the kicked-unitary proposal $q_{t,\omega}(y|x) = |\langle y|W_{t,\omega}|x\rangle|^2$, and every member is reversible with respect to the same π .*

Proof. Each $q_{t,\omega}$ is symmetric (Theorem 2 applies realization-wise after symmetrizing over the time-reversal

involution), so the Metropolis acceptance $a(y, x) = \min(1, \pi_y/\pi_x)$ is q -independent and the off-diagonal $T(y|x) = q(y|x)a(y, x)$ is linear in q . Averaging q averages T off-diagonal; the diagonal absorbs the rest. $\square$

Obstruction 1. For reversible chains sharing π , the Dirichlet form $\mathcal{E}_T(f)$ is linear in T , and $\delta(T) = \inf_f \mathcal{E}_T(f)/\text{Var}_\pi(f)$ is an infimum of linear functionals — concave in T . Hence $\delta(\mathbb{E}[T]) \geq \mathbb{E}[\delta(T_\omega)]$: a mixture is at least as good as the average of its members and *can* beat its best member. The G01 intuition had the inequality backwards. Any true monotonicity would have to use the γ -structure of the mixture weights, not mixture-ness alone.

Obstruction 2. Monotonicity would follow from Peskun ordering if $q_\gamma(y|x)$ were pointwise non-increasing in γ for all $y \neq x$. It is not. At a destructive-interference zero of the coherent walk, $q_0(y|x) = 0$, dephasing strictly *increases* the off-diagonal mass. That is the ENAQT mechanism, alive inside the proposal.

The conjecture cannot even be posed at fixed quench time. Take $\alpha = 1$ (pure mixer $H = \sum_i X_i$), uniform π , and $t = \pi$. Each qubit is an independent rotation with flip probability $\sin^2 t = 0$: the $\gamma = 0$ chain is the identity and $\delta = 0$. Any $\gamma > 0$ gives a positive flip probability. Exact Lindblad at $n = 2$ yields $\delta = 0, 0.145, 0.468, 0.807, 0.990, 0.894$ at $\gamma = 0, 0.05, 0.2, 0.5, 1, 3$. Noise helps enormously at fixed t , because time randomization is itself a dephasing-like resource and a fixed t keeps interference zeros for noise to fill.

B. The solvable corner

The interesting counterexample is window-averaged. Set $\alpha = 1$, uniform π , n qubits, and a per-qubit dephasing rate γ . Each qubit is an independent telegraph-reversed rotation $\theta_i(t) = \int_0^t \sigma_i(s) ds$ with $\sigma_i = \pm 1$ flipping at rate γ . The flip probability is $p_\gamma(t) = (1 - f_\gamma(t))/2$, where $f_\gamma(t) = \mathbb{E}[\cos 2\theta(t)]$ solves

$$f'' + 2\gamma f' + 4f = 0, \quad f(0) = 1, \quad f'(0) = 0, \quad (15)$$

hence $f_\gamma(t) = e^{-\gamma t}[\cos \omega t + (\gamma/\omega) \sin \omega t]$ for $\gamma < 2$ with $\omega = \sqrt{4 - \gamma^2}$. The formula matches the exact Lindblad propagator to six decimals. Laplace transformation at $s = 0$ gives the identity $\int_0^\infty f_\gamma(t) dt = \gamma/2$.

Every kernel P_t , and its t -average, is diagonal in the parity basis χ_S , with eigenvalues $\lambda_S = \mathbb{E}_t[f_\gamma(t)^{|S|}]$. Window-averaging over $t \in [0, T]$ splits odd and even moments. Magnetization modes ($|S| = 1$) have $\mathbb{E}_t[f_0] \rightarrow 0$: time-averaging kills the odd moments, and $\mathbb{E}_t[f_\gamma] \approx \gamma/(2T)$ is small either way. Parity modes ($|S| = 2$) have $\mathbb{E}_t[f_0^2] = \mathbb{E}_t[\cos^2 2t] \rightarrow 1/2$: time-averaging does *not* kill even moments of the coherent oscillation. With $\gamma > 0$, f_γ^2 decays and $\mathbb{E}_t[f_\gamma^2] = O(1/T) \rightarrow 0$. Therefore $\delta(\gamma = 0) = 1/2$, while intermediate γ reaches

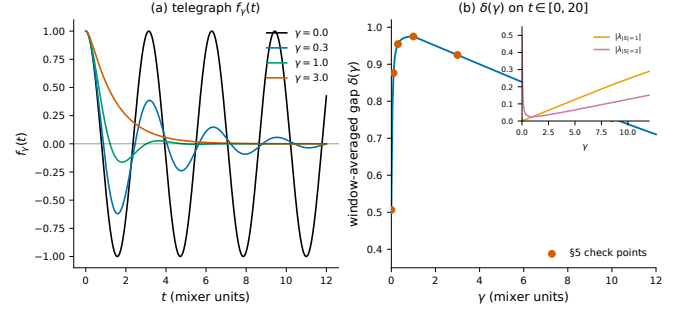


FIG. 2. Solvable corner. (a) Telegraph correlator $f_\gamma(t)$ at representative γ . (b) Window-averaged gap $\delta(\gamma)$ on $t \in [0, 20]$, with an inset of the parity-mode eigenvalues $|\lambda_{|S|=1}|$ versus $|\lambda_{|S|=2}|$. Time-averaging kills the odd sector; only genuine dephasing kills the even sector. The $n = 2$ Lindblad propagator agrees with the formula to 8×10^{-16} .

$\delta \approx 1 - \max(\gamma/(2T), O(1/T))$. The window-averaged gap has an interior maximum.

Measured at $n = 2$ on the window $t\Delta \in [0, 20]$: $\delta = 0.505, 0.874, 0.952, 0.973, 0.923, 0.753$ at $\gamma = 0, 0.1, 0.3, 1, 3, 10$, matching the formula ($0.505 \leftrightarrow 1/2$; $0.753 \leftrightarrow 1 - 10/40$). Conjecture v1 is refuted. The peak $0.5 \rightarrow 0.97$ is the number we claim for this corner.

Time randomization and dephasing are both phase randomizers. Time-averaging kills phases only linearly (odd moments). Coherent correlations that survive in even moments — parity observables — are killed only by genuine decoherence. Noise *can* help mixing, by destroying coherent structure that the time window cannot reach.

That interior-optimum phenomenon is not ours. Kendon and Tregenna [4] found numerically that small decoherence enhances discrete-time walks on the line, cycle, and hypercube, with an optimal rate $p \cdot T \approx 2.6$ –5. Fedichkin, Solenov, and Tamon [5] gave an analytic interior optimum on cycles. On the hypercube itself, Alagic and Russell [6] identified a decoherence threshold for linear instantaneous mixing, and Drezgich *et al.* [7] characterized continuous-time mixing versus Markovian decoherence rate and axis, with an optimal $\gamma/\Delta \approx 1$ –5 obtained from the same non-interacting-qubit factorization used here. Richter [14, 15] already framed decoherent walks as MCMC, for uniform targets and without a Metropolis filter. What is new in this corner is the exact window-averaged telegraph solution, the odd-versus-even-moment mechanism that separates Layden-style time randomization from genuine dephasing, and the embedding of that analysis inside an exact MH sampler. Prior hypercube results concern the walk’s own mixing to uniform.

A finer scan on the e01 chain at $n = 4$ reconciles the pilot with the corner. Interior peaks exist in every cell; their size is controlled by α . At $\alpha = 0.3$ and 0.5 the bumps are 0–8% over $\gamma = 0$ — invisible on G01’s coarse grid at $n = 6$, so the pilot’s “12/12 monotone” is correct for its cells and is not a law. At $\alpha = 1.0$, in the corner’s

neighborhood, peaks reach a factor of two (e.g. $T = 0.3$: $\delta(0) = 0.049 \rightarrow \delta(0.3\Delta) = 0.100$; $T = 1.0$: $0.065 \rightarrow 0.109$). In all fifteen cells the peak stayed below the best classical baseline (single-flip gap 0.17–0.25; uniform-flip up to 0.55 at high T). Noise helped the walk only where the walk was not worth using.

C. The envelope law

Envelope law. For every target/temperature cell,

$$\max_{\gamma>0} \delta(\gamma) \leq \max(\delta(\gamma=0), \delta_{\text{SF}}, \delta_{\text{UF}}), \quad (16)$$

where δ_{SF} and δ_{UF} are the exact gaps of Metropolized single-flip and uniform-flip. Equivalently: wherever the quantum kernel has an advantage, dephasing only erodes it; wherever dephasing helps, a classical kernel was already better.

The law is not a theorem. A plausible route is that the dephased kernel lies in the convex hull of phase-kicked quench kernels, and that this hull’s gap envelope is attained on the boundary {coherent quenches} $\cup$ {Zeno/classical limit}. That convex-hull conjecture is stated, not proved. The paper’s evidence is a pre-registered test, written after the theory pass and before any of the fifty runs. The registered statement includes a 1% slack $\varepsilon = 0.01$ for propagator discretization; a cell is an L1 violation if its peak exceeds $(1 + \varepsilon)$ times the envelope, and a violation $\geq 5\%$ would have refuted the law.

The test draws fifty cells from a seeded rng (master seed 20260818): target class uniform on {e01-chain, SK, RFIM} with a fresh instance, n uniform on {4, 5, 6}, T log-uniform on [0.05, 10], α uniform on [0.05, 1.0]. The γ grid is $\{0\} \cup \text{logspace}(-2, 2, 9)$ in units of Δ_3 , the spectral width of H . The kernel is the same window-averaged family as G01 ($t\Delta_3 \in [2, 12]$, twelve time points). The metric is the exact MH gap versus the enumerated target.

Result. The law holds in 49/50 cells. One disclosed exception, cell 46, is a +1.17% instance-specific micro-bump. Interior peaks occur in 20/50 cells — the nuance is generic. Four cells are advantage cells ($\delta(0)$ beats both classical baselines), all of them low- T SK or RFIM with $\alpha \in [0.63, 0.86]$ and $T \in [0.05, 0.1]$; in those four, the γ -curve is monotone-eroding beyond $\gamma \approx 0.03\Delta$ (sub-2% instance-specific micro-bumps can occur below that). The median peak-to-envelope margin is -72.5% . No cell approached the 5% refutation threshold.

Cell 46 is an SK instance (seed 46), $n = 5$, $T = 0.085$, $\alpha = 0.63$ — itself an advantage cell (quench gap 0.0895 versus single-flip 1.3×10^{-6}). The curve is 0.0895, 0.0906, 0.0815, 0.0203 at $\gamma/\Delta_3 = 0, 0.01, 0.0316, 0.1$, peaking at $\gamma = 0.01\Delta_3$ with margin +1.17% over $\gamma = 0$. A finer γ grid maxes at +1.65% ($\gamma \approx 0.02\Delta_3$, twelve time points) and survives doubling the time-point count (+1.2% at twenty-four points): it is

a real property of this instance, not discretization noise. A fresh SK instance (seed 146) at the identical (n, T, α) is purely monotone. The bump lives at $\gamma \sim 0.01\text{--}0.03\Delta_3$, three orders below the effect H1 needed, and it requires γ -control precision that makes it useless as a resource.

Orfi and Sels prove that *any* unital quantum proposal — dephasing included — has no speedup over classical sampling on their marked-item worst case [8]. That bound already covers our whole dephased family on their adversarial instance, against the uniform baseline. The envelope law is complementary: per-instance, family-wide, quantitative, on typical Ising/SK/RFIM cells, against the full baseline envelope, with γ -resolved erosion. Theirs is worst-case impossibility; ours is a typical-instance accounting.

D. Advantage cells: the G01 picture

The four G05 advantage cells are small- n and sparse. The advantage *regime* — low T , a coherent quench that already beats single-flip, $\alpha \leq 0.5$ — is the G01 pilot: $n = 6$, e01-chain and SK, $T \in \{0.1, 0.3, 1.0\}$, $\alpha \in \{0.05, 0.15, 0.3, 0.5\}$, window $t\Delta \in [2, 12]$, exact gap versus $\gamma/\Delta_3 \in \{0, 0.03, 0.1, 0.3, 1, 3, 10\}$. All twelve cells are monotone decreasing in γ . The E01-type quantum advantage over single-flip persists at low T and is strictly eroded by dephasing; at $T = 1$ the dephased walk falls below the classical baseline by $\gamma \approx 0.1\Delta$. Representative gaps: chain $T = 0.1$, $\alpha = 0.5$ goes $8.3 \times 10^{-4} \rightarrow 1.4 \times 10^{-4} \rightarrow 2.5 \times 10^{-6}$ from $\gamma = 0$ to 0.1Δ to Δ , against single-flip 3.9×10^{-9} ; SK $T = 0.3$, $\alpha = 0.5$ goes $4.5 \times 10^{-2} \rightarrow 4.7 \times 10^{-3} \rightarrow 1.4 \times 10^{-4}$, against single-flip 4.9×10^{-4} .

An ESS confirmation over the pilot’s twenty-four (T, α) cells (both targets; 50 k steps $\times$ five seeds; split- $\hat{R} \leq 1.05$ gating) agrees with the gap ordering wherever it is measurable. Of 216 kernel-cells, 92 are $\hat{R}$ -gated — frozen or near-frozen chains at large γ , and the slow chains at $T = 0.1$. Of the fourteen cells with at least three non-gated γ points, Spearman $\rho(\text{gap}, \text{ESS/step}) > 0.8$ in twelve and equals 1.0 in nine. Zero cells show an interior ESS peak. No $\gamma > 0$ beats $\max(\gamma = 0, \text{classical})$ on ESS/step or ESS/sec. The one automated flag is a known advantage cell in which the *walk family* beats uniform-flip, as it should; within the family, ESS differences across $\gamma \in \{0, 0.03, 0.1\}$ lie inside one standard error. Figure 3 is the γ -scan: $T = 0.1$ cells are gap-only.

Peaks are common (20/50 in G05; every $n = 4$ cell at fine resolution; the solvable corner). Useful peaks are nonexistent. In advantage cells, dephasing erodes the advantage beyond $\gamma \approx 0.03\Delta$. That is the second pillar, in the form a referee can try to break.

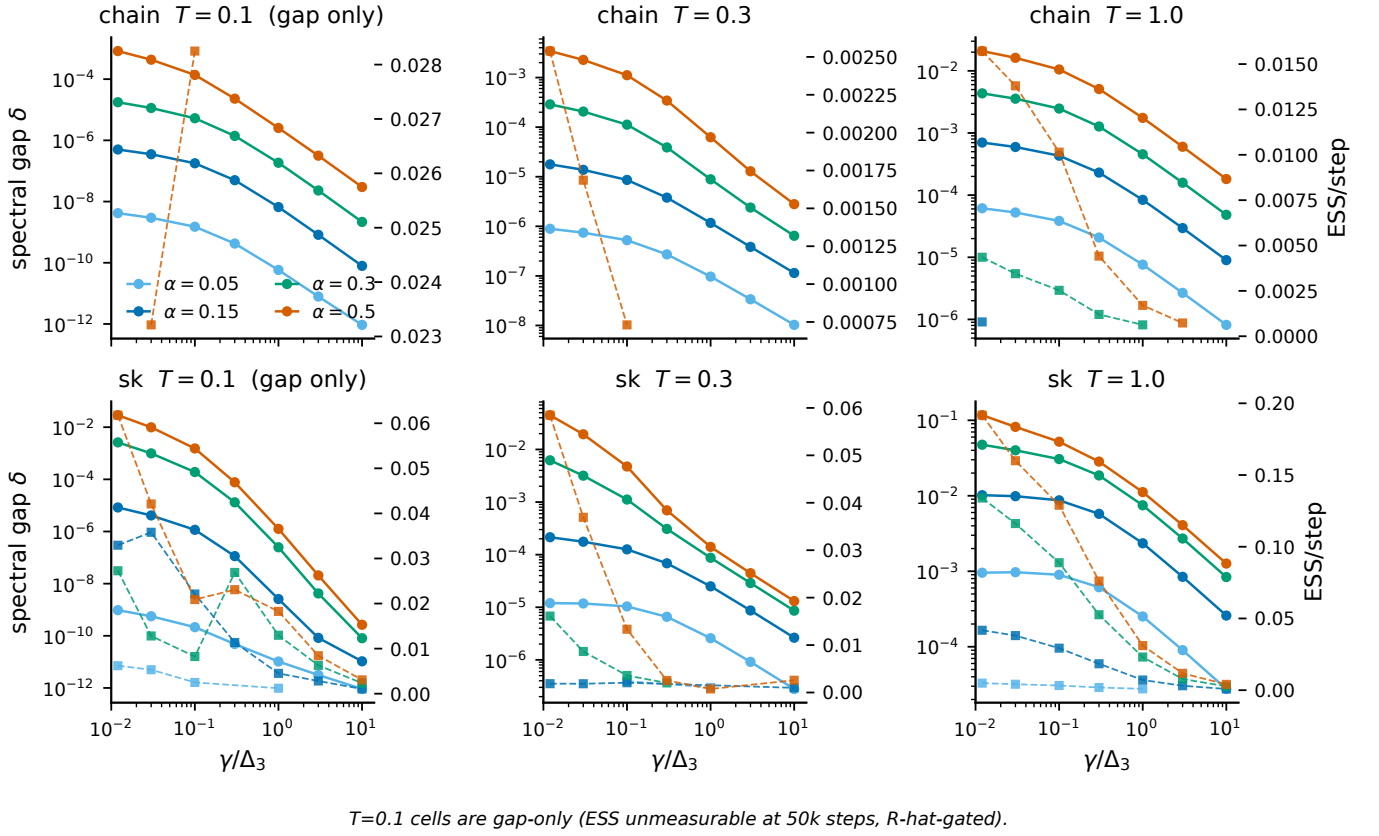


FIG. 3. G01 γ -scan at $n = 6$: exact MH gap versus γ across (T, α) cells, with ESS/step overlaid only on $\hat{R}$ -passable rows. Advantage cells show monotone erosion of the quench. $T = 0.1$ cells are gap-only (ESS unmeasurable at 50k steps, $\hat{R}$ -gated).

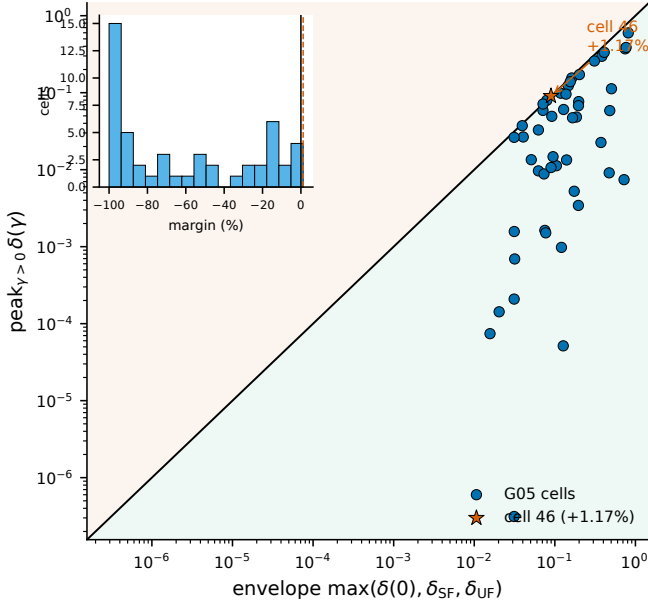


FIG. 4. Envelope law, fifty pre-registered G05 cells: $\text{peak}_{\gamma>0} \delta(\gamma)$ versus $\max(\delta(0), \delta_{\text{SF}}, \delta_{\text{UF}})$. The diagonal is the law boundary. Cell 46 (+1.17%) is marked; the inset is the margin histogram. Cell 19 is a sub- ε +0.087% geometric crossing of the diagonal, not an L1 violation.

V. WHY ENAQT DOES NOT TRANSFER TO SAMPLING

The solvable corner and the G05 peak rate (20/50) show that the ENAQT intuition is not wrong about the *walk*. It is wrong about the *sampler*. The two problems have different objective functions, and the difference is the Metropolis filter.

ENAQT asks for transport to a sink. There is no quality control: every amplitude that arrives counts, whether or not it changed the energy. A little dephasing breaks destructive-interference traps, opens dark pathways, and raises the arrival probability; too much dephasing Zeno-freezes the excitation. The Goldilocks peak is a peak in unfiltered throughput.

MH asks for samples from $\pi \propto e^{-E/T}$. The accept/reject step prices every proposal by its energy change. At low T the filter accepts, essentially, only near-degenerate moves. The coherent quench is valuable *because* its interference concentrates proposal mass on those near-degenerate pairs — the same energy-conserving structure that makes Layden's $\gamma = 0$ kernel beat single-flip by many orders of magnitude on a cold Ising chain. Dephasing is a phase randomizer. It delocalizes, which is the ENAQT benefit, and it broadens the proposal's energy distribution, which is an MH cost.

The filter then rejects the spread. Figure 5 (Sec. VII) is a cartoon of that accounting: a narrow proposal- ΔE peak sitting inside the Metropolis window at $\gamma = 0$, and a broadened peak spilling out of the window at $\gamma > 0$.

The quantitative handle is already in Sec. IV. In every G01 advantage cell the delocalization gain never outruns the acceptance cost: the gap is monotone decreasing in γ , the quench advantage is eroded beyond $\gamma \approx 0.03\Delta$, and at $T = 1$ the dephased walk has fallen below the classical baseline by $\gamma \approx 0.1\Delta$. The pattern is the same in the four G05 advantage cells. The $\alpha = 1$ solvable corner is the complementary check. There the classical energy E is flat, π is uniform, and every proposal is energy-preserving. The Metropolis filter is idle. Noise is then free to help the walk’s own mixing — and it does, $0.5 \rightarrow 0.97$ — but uniform-flip already has gap 1, so the peak is sub-classical by construction. Where the filter is off, ENAQT wins the walk-versus-walk comparison and loses the kernel-versus-envelope comparison. Where the filter is on, ENAQT loses both.

Two distinctions keep the bookkeeping honest. First, time-averaging is also a phase randomizer, but it is not a substitute for dephasing. The window kills odd moments and leaves even-moment (parity) coherences; only genuine noise kills those (Sec. IV). Time-averaging does *not* smear energy on a given realization — each trajectory still evolves under a real Hamiltonian — so it does not pay the MH penalty that dephasing pays. Second, the walk literature’s noise-beats-classical results compare a decohered walk to the classical walk on the same sparse graph, toward a uniform target, with no acceptance filter. That is a different contest than the one the envelope law runs. The Metropolis filter is precisely what flips the verdict: walk-versus-walk, uniform target, no filter, and noise can win; kernel-versus-envelope, Gibbs target, MH filter, and noise never does.

Nothing in this section is yet a design rule for future kernels. The observations fit a picture in which the quench’s value is energy-conserving interference and dephasing spends that value. Whether a scalar “energy concentration” predicts advantage across kernels and targets is stated as a conjecture in Sec. VII and then put to an exact-tier test in Sec. VII A, where the scalar form fails and a concentration-at-distance refinement survives.

VI. BENCHMARKING PROTOCOL

Quantum-algorithm benchmarks are easy to fool. The rules below are the instrument, not the appendix. They are the TunnelVision exactness harness, applied here to a noise-as-resource hypothesis and offered as a reusable method.

Exactness boundary. Proposal kernels never see target log-probabilities. Nothing quantum sits on the accept side of MH. A kernel returns a triple $(y, \log q_{\text{fwd}}, \log q_{\text{rev}})$; for every transpose-closed channel the two logs are identically zero by Theorems 1 and 2, but the interface does

not get to assume that. The G01/G05 runners consume the triple and never a π -lookup from the kernel.

Two clocks. Every experiment reports ESS per step and ESS per second. A kernel that wins per step and loses wall-clock by orders of magnitude is a negative result at the system level. Spectral gap, where the $2^n \times 2^n$ kernel can be built, is a third number, not a substitute.

Mixing gate. Split- $\hat{R} \leq 1.05$ gates a chain before its ESS is believed. The gate exists because frozen chains produce garbage-high ESS on a near-constant series — the pathology that appeared in G05 at $\gamma = 31.6\Delta$ and in 92 of 216 G01 kernel-cells. Those cells are excluded, not averaged away.

Enumerated truth. At n small enough to enumerate, we report error against the exact π (and the exact gap) rather than against a surrogate. No translation layer: the walk graph *is* the spin system.

Pre-registration. Kill criteria are written before data. G01’s H1 rule — an interior γ must beat both $\gamma = 0$ and the largest γ outside ± 2 SE — was fixed in the runner. G05’s envelope law, the 1% slack, the 5% refutation threshold, and the “holds with noted exceptions” band for one or two sub-5% violations were fixed in the pre-registration before the fifty-cell runner existed. The originally tempting law (monotone gap for all targets) was *not* registered: the theory pass had already refuted it, and pre-registering a known-false claim would have been theater.

Negatives at full strength. A failed hypothesis is a result. H1 is reported as killed, not as “more work needed.” Cell 46 is reported in full, not absorbed into a mean. The un- $\sqrt{}$ stationary bound is flagged as empirical. The interior-optimum mixing phenomenon is cited as known (Sec. VIII); we claim only the analysis we own.

The protocol earned its keep twice. It killed H1 on the G01 pilot: twelve of twelve cells monotone in the gap, zero ESS peaks under $\hat{R}$ gating, envelope intact on both clocks. It then caught the overreach of conjecture v1. The theory pass produced interior peaks at $\alpha = 1$ and on a fine $n = 4$ grid; registering monotonicity at that point would have been a pre-registered falsehood. The corrected, pre-registered claim is the envelope law, tested on fifty cells the theory pass had never seen. That sequence — kill fast, then refuse to promote a refuted conjecture — is the method contribution. H2 and H3 are moot as stated: no useful γ^* exists, so a matching-condition study and a hardware-idle test do not proceed.

A. Extraction pricing, and an amortized architecture killed by its own registration (run P01)

The two-clocks rule has a hardware-side sharpening. Present devices deliver 10^2 – 10^4 useful shots per second; a classical sampler executes 10^6 – 10^8 proposals per second. A hybrid that consumes one device shot per MCMC step therefore pays a 10^4 – $10^6 \times$ throughput tax before any per-proposal quality enters, while the measured per-

proposal edge of the best quantum kernel in our zoo is $1.8\text{--}26\times$ (Sec. VII A). On any target whose π -evaluation is cheap relative to shot latency ($\sim 10^{-2}\text{ s}$) — the scope matters: an expensive- π target would make per-evaluation and wall-clock currency coincide — no per-step-device hybrid can show an honest ESS-per-second advantage until shot rates improve by roughly three orders of magnitude. This is the general I/O argument of Hoefler *et al.* [16] applied as a design selection rule for sampling, and recent per-step proposals [17, 18] report spectral-gap or per-step gains with no wall-clock metric — precisely the accounting the protocol above forbids. We suggest benchmarks report, alongside the two clocks, the extraction cost $B_{\text{ext}} = \text{device shots consumed/ESS}$.

One architecture escapes the tax structurally: pay the device *once*, offline, to mine a fixed library of low-energy “atom” states, then run a purely classical exact chain whose mixture kernel darts into the frozen library (Tjelmeland–Hegstad mode jumping [19]; precomputed-library darting is classical prior art in continuous spaces [20, 21] and per-step-optimization variants exist for discrete ones [22]). The device cost amortizes; B_{ext} falls with chain length. We raced this architecture under pre-registration (run P01, protocol and data with the manuscript source), with two kill criteria fixed before any run: K1, classical miners (solver member-enumeration without completeness proofs, parallel-tempering restarts, greedy descent with kicks) match the simulated quench miner’s mode coverage at matched builder wall-clock — quantum arm dead; K2, darting gain below $2\times$ over a parallel-tempering baseline on targets pre-screened for measured classical metastability — architecture dead on this class. A portal-acceptance gate (energy-matched atoms must accept at $\gtrsim 0.2$) passed at $n \leq 20$, so the race ran.

Both kills fired. At $n \leq 20$ against enumerated truth, the best classical miner matched or beat the quench miner’s mass-weighted mode coverage in 24 of 24 cells at $n \geq 16$ — with the quench arm overrunning its builder budget $3\text{--}50\times$ (one statevector evolution at $n = 20$ costs $30\text{--}100\text{ s}$). At $n = 50\text{--}200$, six of twelve cells survived the metastability screen (the others mix under plain single-flip MH — racing there would manufacture a trivial K2 failure), and on the survivors the darting gain was $0.46\text{--}1.75\times$, never $2\times$, with two measured mechanisms: solver member-finding collapses from hundreds of atoms to $1\text{--}3$ per 30-second budget on dense instances at $n \geq 50$ (bounding the scale reach of the member-finding speed reported in Sec. VII A), and portal acceptance falls to $\lesssim 0.01$ because energy matching at $\delta E \lesssim T$ does not transfer upward in n . Whether quantum hardware would out-mine classical miners at $n \geq 50$ is not decidable in simulation (statevector caps near $n = 28$; tensor-network truncation corrupts the shell physics being measured) and we make no claim either way. The architecture that structurally escapes the extraction tax therefore fails on dense spin glasses for classical reasons — which is the benchmark working as designed: the registered negative,

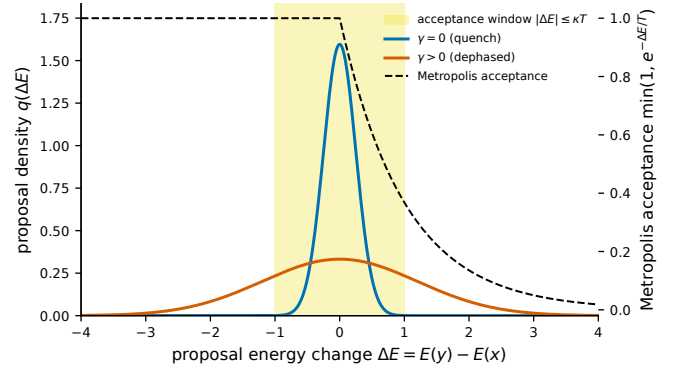


FIG. 5. Proposal energy change $\Delta E = E(y) - E(x)$ at $\gamma = 0$ (narrow, sitting inside the Metropolis window) versus $\gamma > 0$ (broadened, spilling out of the window). The overlaid curve is the Metropolis acceptance $\min(1, e^{-\Delta E/T})$. Energy concentration $C(q)$ is a proposed predictor of advantage, not a theorem of this paper.

its screen, and B_{ext} are reusable as protocol pieces, and offline-batch device reuse remains live only where the consumer is an estimator rather than a chain [23].

VII. ENERGY CONCENTRATION AS A PREDICTOR: CONJECTURE, TEST, REFINEMENT

The observations of Secs. IV and V fit a single accounting: the quench is valuable when its proposals sit on near-degenerate pairs, and dephasing spends that value by broadening proposal energy. We record that accounting as a conjecture, not as a theorem of this paper.

Conjecture (energy concentration). The advantage of a proposal q over trivial classical kernels is predicted by the proposal mass on near-degenerate pairs,

$$C(q) = \sum_{x,y} \pi(x) q(y|x) \mathbf{1}[|E(y) - E(x)| \leq \kappa T], \quad (17)$$

for an $O(1)$ window κ . Kernels should be engineered for energy concentration, not for delocalization.

Everything already measured is consistent with $C(q)$ and does not prove it. Dephasing lowers C (broadened proposal energy) and erodes advantage (G01, G05 L2(b)). Time-averaging preserves C , because energy conservation is per-realization of a real Hamiltonian, and does not erode advantage the way dephasing does. The $\alpha = 1$ corner has flat E , so C is identically 1 and noise can only help sub-classically — which is what the telegraph solution shows. Figure 5 is the cartoon of that split, not a fit to data.

A. The test, executed: the scalar fails; a refined predictor holds

That test has now been run, on a kernel zoo wide enough to break the conjecture where it is breakable (run M01; data with the manuscript source). For fourteen exact-tier kernels — the coherent quench; single-flip, add–delete–swap, and uniform-flip; a Creutz-demon microcanonical kernel [24] at two demon temperatures; discrete Langevin proposals [25] at three informedness levels, including the uninformed random-mask limit; two “soft-spin” variants that flip weak-local-field sites preferentially; and an oracle *shell kernel*, uniform on $S_\varepsilon(x) = \{y \neq x : |E(y) - E(x)| \leq \varepsilon\}$ with the $|S_\varepsilon|$ Hastings correction, at three widths — we computed $C(q)$ ($\kappa = 2$) and the exact MH gap on six all-to-all spin-glass instances with random fields ($n = 8, 10$; seeds 1–3) at $T \in \{1, 0.3, 0.1\}$: 18 cells, 252 kernel evaluations.

The scalar conjecture fails. Across the eleven implementable kernels the per-cell Spearman correlation between $C(q)$ and the gap has median $\rho_s = -0.07$ (range -0.70 to $+0.97$ across the 18 cells). The counterexamples are systematic, not noise: informed local kernels — discrete Langevin at full informedness, soft-spin at strong bias, the cold demon — post the *highest* energy concentrations in a cell while sitting up to ten orders of magnitude below the quench in gap. They concentrate energy the cheap way, by proposing single-site or downhill moves, and the same information that concentrates them makes them greedy and local. Figure 6(a) is that scatter.

Two amendments repair the predictor, and both are dictated by the data rather than fitted to it. First, the window that matters is set by the coupling scale, not the temperature: quench and shell-oracle gaps are T -independent from $T = 1$ to $T = 0.1$ while a κT window empties. Second, concentration must be achieved *at distance*. Define

$$C_{\text{far}}(q) = \sum_{d_H(x,y) \geq 3} \pi(x) q(y|x) \mathbf{1}[|E(y) - E(x)| \leq \varepsilon], \quad (18)$$

with ε of order the coupling scale ($\varepsilon = 2$ here, $J_{ij} \sim \mathcal{N}(0, 1)$). The per-cell correlation jumps to median $\rho_s = +0.84$ (minimum $+0.26$), and the quench is the top-ranked implementable kernel by C_{far} in 16 of 18 cells (mean rank 1.17) — exactly where it is the top-ranked kernel by gap [Fig. 6(b)].

The oracle closes the loop from correlation to mechanism. A kernel with *only* the shell property — uniform on $S_\varepsilon(x)$, knowing nothing about the walk, the mixer, or coherence — matches or beats the quench gap on every instance at $\varepsilon \approx 2$ –4 (e.g. $n = 10$, seed 3, $T = 0.1$: shell 0.19 vs. quench 0.045; $n = 8$, seed 1: 0.15–0.19 vs. 0.085–0.090), with the one caveat that $\varepsilon \lesssim 1$ can disconnect the state space (gap 0; connectivity is a separate necessary condition the scalar does not see). The quench’s value on these targets is that it *is* an approximate global shell sampler.

The qualitative half of this identification has antecedents in the qe-MCMC literature, and we flag them explicitly. Ferguson and Wallden [26] state, as the design heuristic behind their coarse-graining, that proposals which maximize Hamming distance while minimizing energy difference are the ones likely to be accepted, and engineer their scheme to retain both properties. Arai and Kadowaki [17] measure exactly this signature — large d_H jointly with small $|\Delta E|$ — for quantum-annealing proposals on spin glasses and identify it as the acceptance source. Cao *et al.* [27] note in passing that the quench induces long-range transitions between near-degenerate states. What those works assert or observe qualitatively, the zoo makes quantitative and falsifiable: C_{far} is a measured predictor with a refuted scalar rival, and the oracle shows the shell property alone — no interference, no mixer, no coherence — suffices, which none of the antecedents constructs or tests.

What no kernel in the zoo achieves is the conjunction cheaply. Local information buys concentration only at $d_H = 1$, and adding more of it (discrete Langevin informedness, soft-spin bias, demon budget) moves kernels monotonically *away* from the quench corner of Fig. 6(b). Distance without information (uniform, random-mask) forfeits concentration. Selecting shell candidates from random masks by multiple-try Metropolis [28] matches the quench per accepted step in sampled runs but spends $2K - 1 = 15$ target evaluations per step doing it.

The classical literature approaches the conjunction from two directions outside our zoo, and neither closes it. Houdayer’s isoenergetic cluster moves [29], extended to arbitrary dimension by Zhu, Ochoa, and Katzgraber [30], are $O(n)$, rejection-free, and global in Hamming distance — but they conserve the joint energy of a replica *pair* rather than the energy of the updated chain, propose only within the instantaneous two-replica overlap structure rather than uniformly on a shell, and degenerate as clusters percolate on dense frustrated graphs such as these. We have measured all three failure modes on the M01 instances (run M02, data with the manuscript source): on these all-to-all targets the faithful cluster move is exactly the replica swap — a statistical no-op — restricted clusters forfeit pair-energy conservation and post zero pair-Metropolis acceptance at $T = 0.1$, the two-replica disagreement set is empty in 52–100% of stationary draws at $T = 0.1$, and mixing cluster moves into single-flip at equal weight never lifts the exact two-replica gap above the mixture-dilution floor (0.50 – $0.51 \times$ the single-flip pair baseline in every resolvable cell). The equi-energy sampler of Kou, Zhou, and Wong [31] jumps within energy rings at arbitrary Hamming distance — statistics has wanted this oracle for two decades — but draws its jump targets from stored past states of auxiliary higher-temperature chains, so shell samples are amortized from prior mixing, not generated fresh. Simulating the quench itself with matrix-product states, as Christmann *et al.* [32] propose, reaches the shell at polynomial but distinctly superlinear cost per proposal, with the re-

ported advantage surviving even unconverged bond dimension.

Whether a classical kernel can sample the energy shell at distance *de novo* with $O(n)$ – $O(n^2)$ work per proposal is, after this test, the sharpest open question this program leaves; at $n \leq 10$ nothing we constructed does, and the quench’s per-proposal edge over the best implementable classical kernel in each cell runs from $1.8\times$ to $26\times$ in gap (median $11\times$), and the quench is the top-ranked kernel by gap in all 18 cells. (Per wall-clock second on present hardware that edge is inverted by four orders of magnitude of shot cost; the point here is mechanism, not deployment.)

The oracle is nonetheless NP-cheap, and we have measured what that costs — with one instructive false start. Encoding the shell as integer pseudo-Boolean constraints and sampling it with random parity constraints, in the style of hashing-based near-uniform SAT sampling [33–35], yields *de novo* shell samples that agree exactly with enumeration on the integerized Hamiltonian and are empirically uniform (total variation ≤ 0.03 from uniform on every enumerable instance; run M03, data with the manuscript source). Along a temperature ladder at $n = 20$ (run M05) the cost of this CNF-based route rises steeply with depth: shells around states equilibrated above T_c ($\sim 10^5$ members) sample in $\sim 10^{-1}$ s, while below T_c the shells shrink to 10^3 – 10^0 members and CryptoMiniSat, UniGen, and ApproxMC all need 10^2 – 10^3 s or time out. That wall is largely an artifact of the CNF translation: a pseudo-Boolean solver that takes the energy window natively (Exact) enumerates the same deep shells in 10^{-1} – 10^1 s, 10 – $3,500\times$ faster on identical states. Finding members is not the expensive part; an *exact* proposal must also certify that no further member exists (for the $|S_\varepsilon(x)|/|S_\varepsilon(y)|$ Hastings ratio, or to draw uniformly), and that completeness proof is what scales. (Member-finding being cheap is itself a bounded claim: it is measured here at $n \leq 32$, and run P01 finds it fails by $n = 50$ on dense instances at this window — Sec. VIA.) With the native solver and correctly equilibrated low-energy states (run M06), an exactly uniform deep-shell proposal costs 1–4 s at $n = 16$, 30–50 s at $n = 20$, and is censored at our 300 s per-call timeout in eleven of twelve deep cells at $n = 24$ – 32 — growth of at least an order of magnitude per four spins — against $\sim 10^{-2}$ s per quench proposal on hardware, flat in n at these sizes. The gap is therefore three and a half orders at $n = 20$ and at least four and a half, widening, from $n = 24$. Two facts survive tool choice: the deep shells the quench targets are tiny at this window, and every classical route to them has to *decide* near-empty integer windows — a completeness proof a physical sampler never has to produce, and one that no cheap kernel of Fig. 6 attempts at all.

Refined conjecture (concentration at distance). Advantage over trivial kernels is predicted by $C_{\text{far}}(q)$ — proposal mass that is simultaneously near-degenerate at the coupling scale and long-range — subject to shell connectivity. The scalar $C(q)$ alone is refuted as a design

objective by the informed-local counterexamples above.

Three escapes from G01 were not pursued, and we do not want them read as hidden rescues of H1. Site-dependent rates γ_i could *shape* where proposals land rather than how coherent they are. Noise in the basis of a frustrated subspace, rather than the computational basis, is a different channel, still transpose-closed if the jumps are real-symmetric (Theorem 1) but not the family G01/G05 scanned. Hitting-time objectives — first arrival at a ground-state set — restore a sink, and are optimization, not sampling. Each is a new hypothesis. None of them, if true, would change Theorems 1–4 or the envelope law as stated for uniform computational-basis dephasing inside exact MH.

VIII. RELATED WORK

We group neighbors in the order a reader should meet them: the decohered-walk mixing literature that already owns the interior-optimum phenomenon; the quantum-enhanced MCMC line we sit on; and ENAQT, whose objective we do not share.

A. Decohered-walk mixing — the known phenomenon

Fastest mixing at intermediate decoherence is established walk literature, including on the hypercube that is our $\alpha = 1$ graph.

Kendon and Tregenna [4] showed numerically that small decoherence enhances discrete-time walks on the line, cycle, and hypercube, with an optimal rate $p \cdot T \approx 2.6$ – 5 and a mixing time on the cycle below the classical value. Fedichkin, Solenov, and Tamon [5] gave the analytic counterpart on cycles: mixing time improves linearly in the decoherence rate at small rates, degrades linearly at large rates, and has a unique interior optimum. Abal *et al.* [36] found a mixing-time minimum at broken-link probability $p \approx 0.1$ for a discrete-time hypercube walk.

On the continuous-time hypercube, Alagic and Russell [6] identified a decoherence threshold below which linear instantaneous mixing survives, with classical $\Theta(n \log n)$ behavior and Zeno retardation beyond it. Drezgich *et al.* [7] gave a complete characterization versus Markovian decoherence rate *and* axis: a finite optimal rate $\gamma/\Delta \approx 1$ – 5 for almost all decoherence axes (none for axes in the x – y plane), derived from the same non-interacting-qubit factorization our telegraph solution uses. Those results are fixed- t instantaneous mixing to uniform; they have no window average, no Metropolis filter, and no classical-envelope comparison.

Richter [14, 15] is the closest in spirit: decoherent walks always mix; mixing is robust to any smooth decoherence; the framing is already MCMC/sampling; threshold mixing is proved for the hypercube $\mathbb{Z}_2^n$; a $\sqrt{\delta}$ quantum

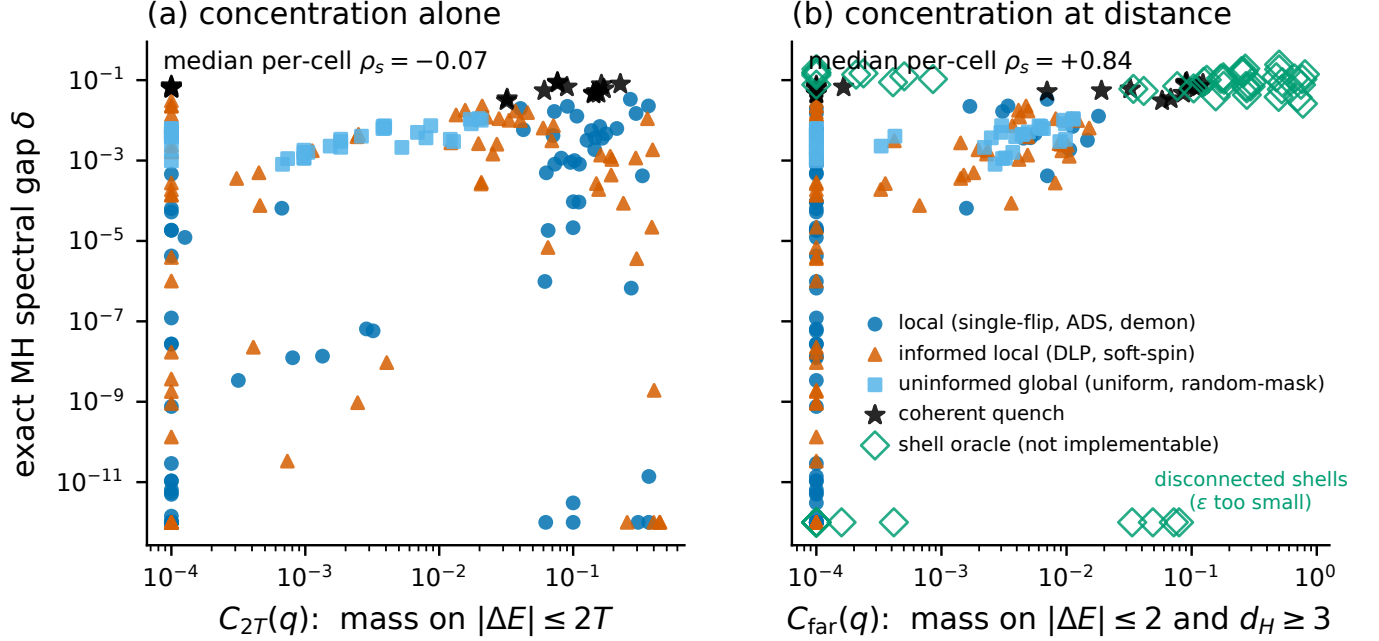


FIG. 6. The Sec. VII test, executed (M01: 14 kernels $\times$ 6 spin-glass instances, $n = 8, 10 \times T \in \{1, 0.3, 0.1\}$; exact MH spectral gap throughout; gaps and concentrations clipped at 10^{-12} and 10^{-4} for display). (a) The C8 scalar: energy concentration $C(q)$ with window $2T$ against the exact gap, implementable kernels only. Median per-cell Spearman $\rho_s = -0.07$: concentration alone does not predict advantage, and the informed local kernels (triangles) occupy the high- C , low-gap corner. (b) Concentration at distance [Eq. (18)]: median per-cell $\rho_s = +0.84$, the quench (stars) is top-ranked by C_{far} in 16 of 18 cells, and the shell oracle (open diamonds) — uniform on the energy shell, no quantum structure at all — matches or beats the quench everywhere it is connected. Bottom-right diamonds are disconnected small- ϵ shells: connectivity is a separate necessary condition.

speedup of classical mixing is conjectured. The target is uniform; decoherence is repeated measurement; there is no Metropolis filter, no Gibbs target, and no noise-rate optimization against a classical-kernel envelope.

The quantum-stochastic-walk umbrella [37] and the transport-optimal 90%/10% quantum/classical mixture of Caruso *et al.* [38] belong here as framework, but the latter optimizes transport to a sink, not mixing.

We cite this group first and prominently because it is the phenomenon behind our solvable corner. Our claim in that corner is the window-averaged telegraph solution, the odd/even-moment mechanism, and the exact-MH embedding — not the existence of an interior mixing optimum.

B. Quantum-enhanced MCMC

Layden *et al.* [1] is the $\gamma = 0$ limit of our kernel: coherent quench proposals, exact MH, noise treated as a nuisance. Their Supplemental Material already states the convergence condition we formalize — errors are harmless provided they do not break $Q(s'|s) = Q(s|s')$ symmetry on average — with SPAM-twirling mitigation, and notes that depolarizing noise degrades the proposal toward uniform. Theorems 1 and 2 are the systematic channel-level classification of which physical noise satisfies that condi-

tion (transpose-closed Kraus, unitality, T_1 as the unique realistic bias channel). That is a strictly stronger statement, and not a bolt from the blue.

Orfi and Sels [8] prove that *any* unital quantum proposal has no speedup over classical sampling on their marked-item worst case. Dephasing is unital, so their bound covers our whole dephased family on that adversarial instance, against the uniform baseline. Their companion analysis bounds the chain's gap by the inverse participation ratio of classical states in the quench eigenbasis and concludes that useful quenches must be fine-tuned [39], a structural barrier that our C_{far} predictor renders operational. The envelope law is complementary: per-cell, pre-registered, quantitative, on typical Ising/SK/RFIM instances, against the full baseline envelope, with γ -resolved erosion. We cite them as a sibling, not as a special case of us or the reverse.

Follow-ups in the qe-MCMC line have not studied tuned noise. Coarse-grained qe-MCMC [26] says explicitly that the effect of noise “has not been investigated.” The quantum-inspired surrogate of Christmann *et al.* [32] repeats the noise-affects-only-efficiency point without a γ scan. The causal-set application [40] has no noise angle. On the mechanism side the same line contains the qualitative antecedents of Sec. VII A, discussed there: the coarse-graining of Ref. [26] is engineered around large- d_H , small- $|\Delta E|$ proposals; Arai and Kadowaki [17] observe

that signature in quantum-annealing proposals; Cao *et al.* [27] note long-range near-degenerate transitions while pursuing irreversible variants. Relative to all three, what is new in Sec. VII A is the oracle-sufficiency test, the quantitative predictor with its refuted rival, and the classical no-go evidence.

Fault-tolerant Metropolis walks are a different machine model: the walk *is* the chain, not a noisy proposal inside classical MH. We cite Lemieux *et al.* [41], Claudon *et al.* [42], penalised qubitized walks [43], and Incudini and Mazzola [44] in passing.

Dissipative Gibbs samplers [45, 46] engineer a Lindbladian whose fixed point *is* the Gibbs state. That is constructive dissipation, not ambient tuned dephasing inside an exact MH proposal.

C. ENAQT and device-noise sampling

Rebentrost *et al.* [2] and Lloyd and Mohseni [3] are the transport Goldilocks; their citation trees stay in photosynthesis and exciton transport. Digital ENAQT [47, 48] supplies the circuit unravellings we reuse (stochastic-Hamiltonian phase kicks; collision scheme). Those works target transport simulation; they note that converting intrinsic device noise into a programmable stochastic process has not been achieved. D-Wave device-noise sampling [49] is analog and biased, with no exactness layer.

D. Positioning

A fourth neighborhood enters through Sec. VII A: informed proposals for discrete distributions. Locally balanced proposals [50], Gibbs-with-gradients [51], and discrete Langevin proposals [25] use local energy differences to shape single- and multi-site flips; multiple-try Metropolis [28] selects among candidate sets; Creutz’s demon [24] is the deterministic microcanonical ancestor of energy-conserving moves. Isoenergetic cluster moves [29, 30] and equi-energy jumps [31] are the classical moves closest to the shell mechanism itself; Sec. VII A records why neither supplies the cheap de-novo shell generator the open question asks for. These methods are state of the art for the contests they were built for. Our zoo result places them relative to the quench on cold rugged targets: their local information buys energy concentration only at Hamming distance one, which is precisely the corner of Fig. 6 the Metropolis filter does not reward.

What is ours: the first channel-level exactness characterization for noisy walk proposals, upgrading Layden’s SM condition to a classification; the first study of decohered-walk *mixing* results embedded in an exact MH sampler with Gibbs targets; the envelope law as the negative answer to noise-as-resource for *sampling*; and the oracle-sufficiency demonstration that shell sampling alone reproduces the coherent quench’s edge — sharpen-

ing the qualitative large- d_H /small- $|\Delta E|$ picture already present in the qe-MCMC literature [17, 26, 27] — with the concentration-at-distance predictor and the classical no-go evidence of the kernel zoo. The interior-optimum mixing phenomenon itself is 2003–2009 walk literature and is cited as such. The Metropolis filter is what separates the verdicts. Walk-versus-walk, uniform target, no filter: noise can win. Kernel-versus-envelope, Gibbs target, MH filter: noise never wins.

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DATA AVAILABILITY

The certification scripts, figure generators, pre-registration, and numerical results that support the claims of this paper are included with the manuscript source: the R1 and T_1 -bias runners and tests I10–I13; the figure generators under `paper/figures/`; the G05 pre-registration, `results.csv`, and verdict; the G01 ESS confirmation table; the certified R1 and amplitude-damping result files; and the M01 kernel-zoo table (`paper/figures/data/m01_cfar_full.csv`, generated by `tunnelvision/experiments/D01_demon/`, which also carries the demon, multiple-try, discrete-Langevin, and shell-oracle implementations with their exactness tests, and the M02 isoenergetic-cluster measurements (`m02a_icm_cfar.csv`, `m02b_icm_gap.csv`) with the two-replica kernel and its certification tests; the M03 SAT-shell sampler, its pre-registered gates, and its uniformity and wall-clock tables live in `tunnelvision/experiments/M03_shellhash/`; the M05 temperature-ladder and pseudo-Boolean-solver comparison in `tunnelvision/experiments/M05_depth/`, whose results doc also records a state-generator bug that invalidated the depth attribution of the earlier M03/M04 “cold” timings; and the M06 native-solver scan in `tunnelvision/experiments/M06_pbnative/`; the P01 pre-registration, miner race, and results in `tunnelvision/experiments/P01_portal/`; and the sector-support proof note and its Lindblad certification script in `tunnelvision/docs/`). A public archival deposit of both repositories is available at Zenodo, DOI 10.5281/zenodo.22758619.

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